2 – Probabilistic robotics

Advanced Methods for Mapping and Self-localization in Robotics
MPC-MAP

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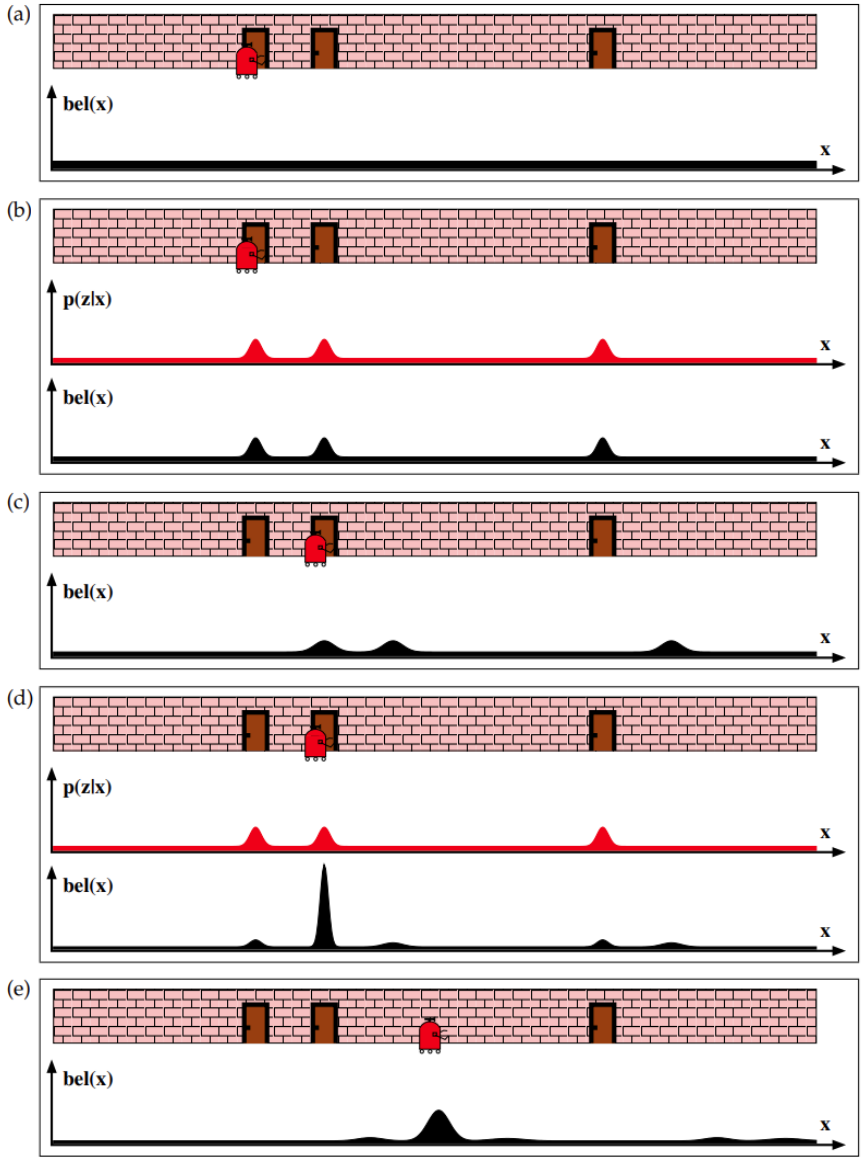
Probabilistic robotics

Key ideas of the probabilistic approach

1D localization example



[1]

I‘ve a map, but I‘ve absolutely no idea where I am.

Let's use my door sensor! It says I'm probably at a door.

Ok, I'm probably at a door.

Let's go on. I'm getting lost, but I still have some ideas where I could be.

Let’s use the sensor again. Great, I'm at another door!

Based on my previous guess and this new information I'm very likely at the second door.

Let's go on. I'm getting little lost again, but I still know where I probably am.

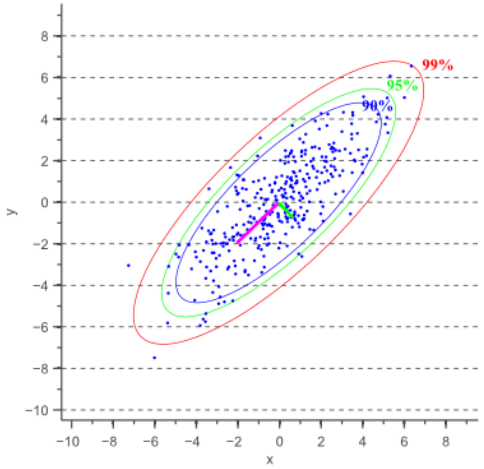
1. http://www.probabilistic-robotics.org/

Probabilistic robotics – the concept

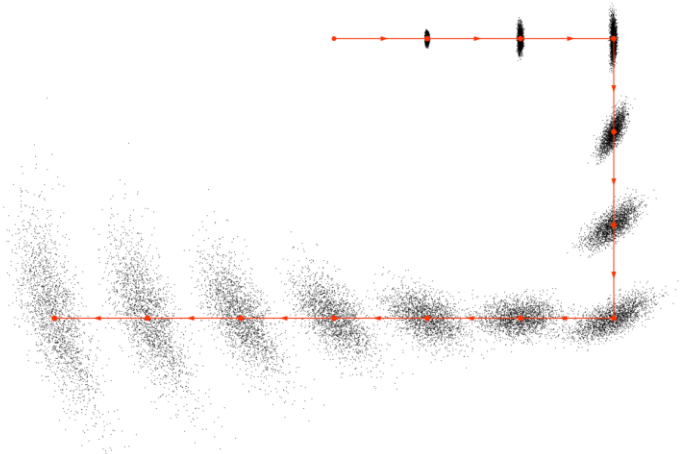
- **Probabilistic approach – nothing is deterministic, all phenomena are stochastic**

- Uncertainty is in both:
  - **Perception** – sensor/measured data are not 100 % reliable; data from all real sensors are more or less noisy
  - **Action** – robot actions/movements are not 100 % reliable; all actions taken differ to some extent from the action desired

- All uncertain phenomena are **represented statistically**



Noisy measurement (e.g., GNSS positioning data) [1]



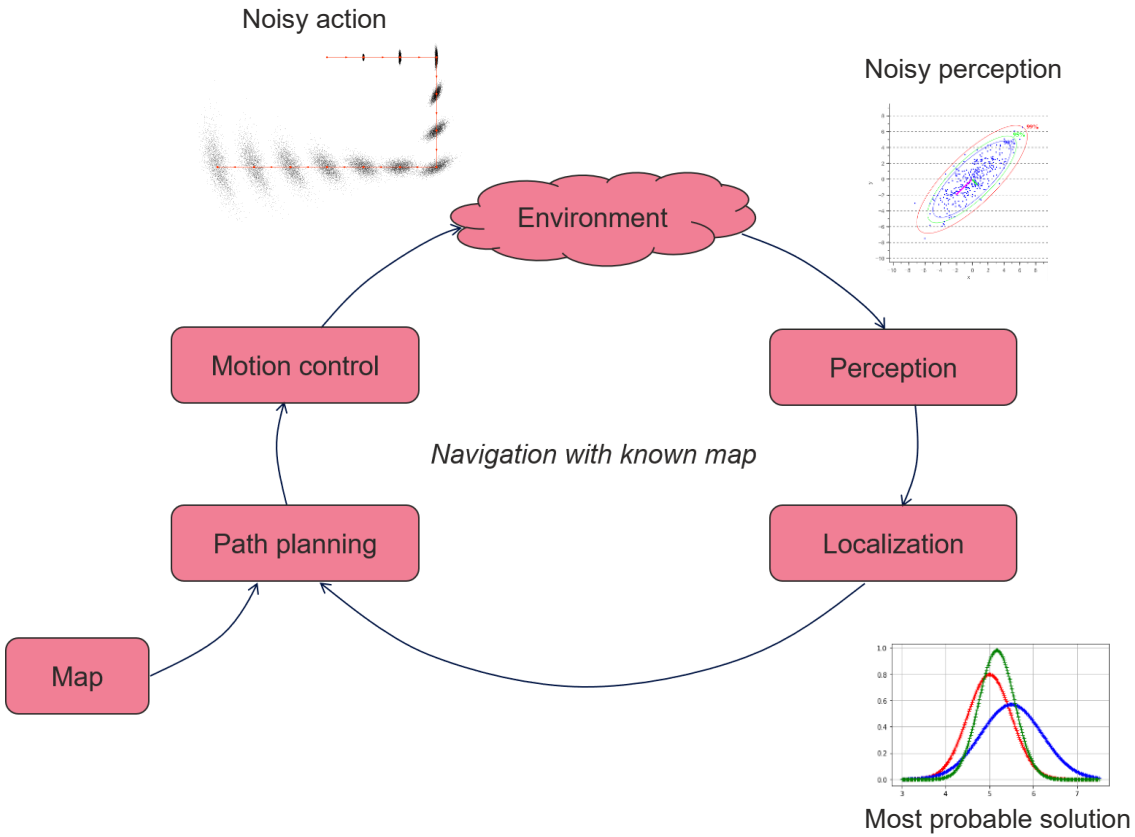
Noisy motion - propagation of error [2]

1. https://www.skyhook.com/blog/best-practices-measure-location-accuracy-geo-positioning-performance-metrics
2. https://en.wikipedia.org/wiki/Monte_Carlo_localization

Probabilistic robotics – the concept

- Robot **navigation** in the probabilistic concept **considers the uncertainties** to provide optimal solution.

- This is especially crucial for the **localization** task – what is the **most probable robot position** according **noisy sensor data**? And considering **known uncertain actions**?



Probability

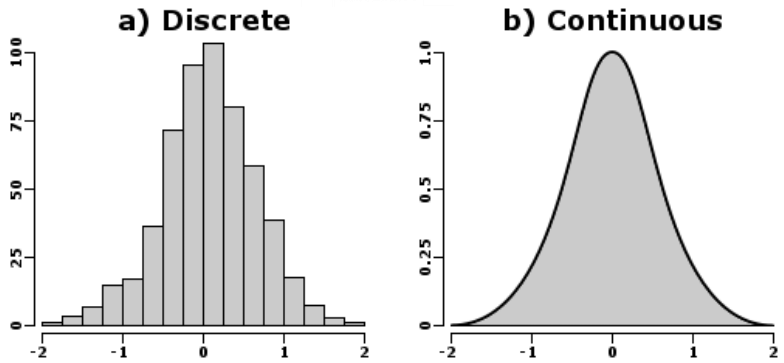
Basic terms and distribution types

Probability

- **Probability** expresses **how likely** a random **event occurs**
  - A number between **0 and 1**; higher value → higher probability

- **Probability distribution**
  - „*Mathematical **function** that gives the **probabilities of occurrence** of different possible **outcomes** for an experiment.*“ [1]
  - „*Mathematical **description of a random phenomenon** in terms of its **sample space** and the **probabilities** of events.*“ [1]

  - **Discrete** space → **probability mass function** (PMF)
  - **Continuous** space → **probability density function** (PDF) or cumulative distribution function (CDF)

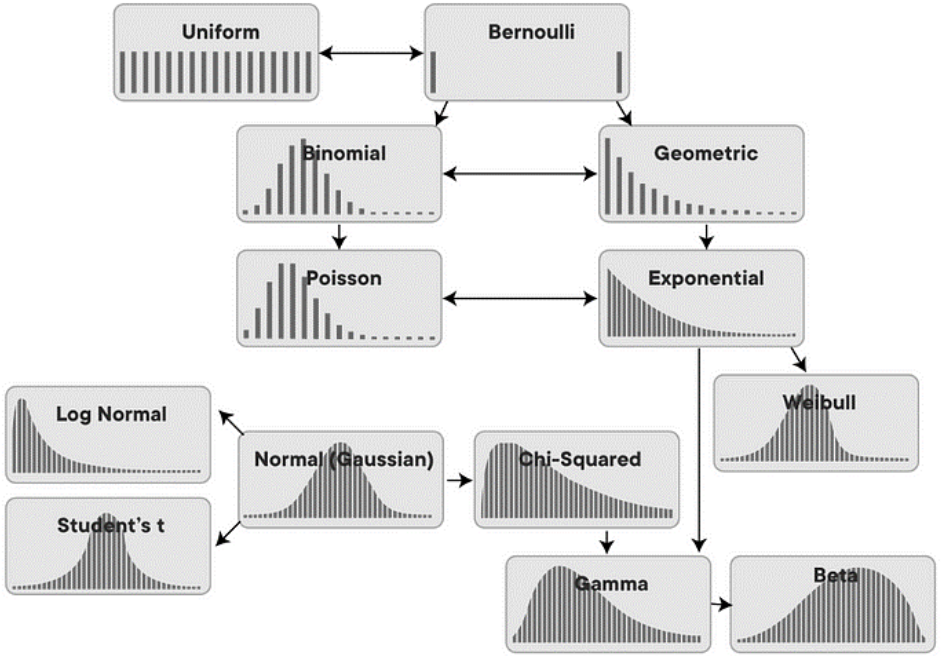


PMF (left) and PDF (right) examples [1]

1. https://medium.com/analytics-vidhya/probability-distributions-444e7babf2e1

Probability mass function (PMF)

- **Probability mass function** (PMF) – probability that a **discrete random variable** is exactly equal to some value
  - $p_X(x) = P(X = x)$
- The probabilities sum up to 1
  - $\sum_x p_x(x) = 1$
- Values are non-zero, positive
  - $p_X(x) \geq 0$

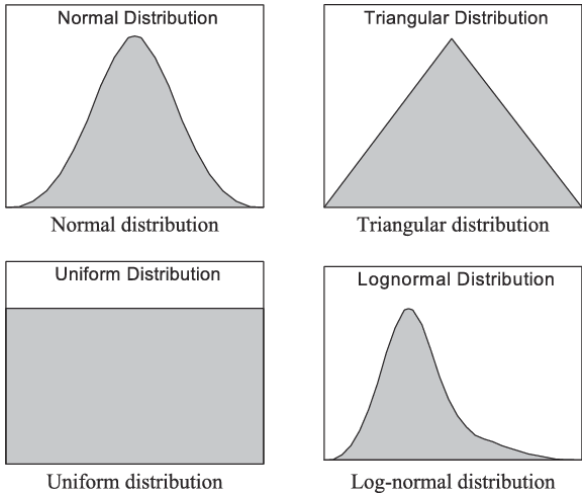


Various probability mass functions (PMF) [1]

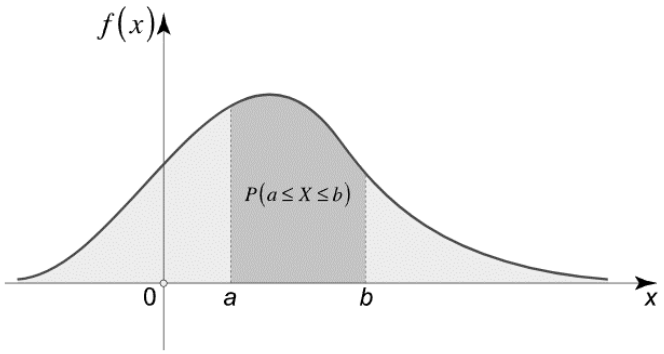
1. https://medium.com/@amanatulla1606/understanding-probability-distributions-in-statistics-and-its-application-in-machine-learning-e62751e83c39

Probability density function (PDF)

- **Probability density function** (PDF) – a function f_X describing the infinitesimal probability of any given value of a **continuous random variable** X
- The **probability** P of a **particular value** equals **0**
  - $P(X) = f_X(x) = 0$
- The **probability** of a **particular range** of value equals to the **integral** of the PDF over the range
  - The area between the PDF and horizontal axis over the range
  - $P(a \leq X \leq b) = \int_a^b f_X(x)\, dx$
- PDF is nonnegative everywhere, integrates to 1
  - $P(-\infty \leq X \leq \infty) = \int_{-\infty}^{\infty} f_X(x)\, dx = 1$



PDF examples [1]

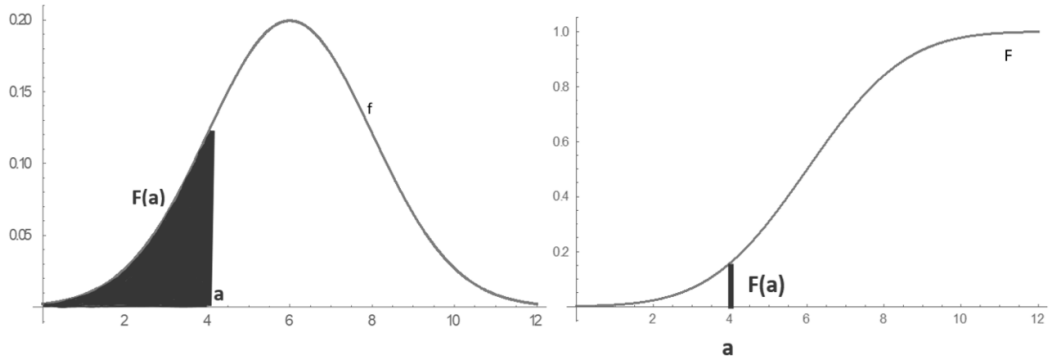


Probability equals to the area under the PDF [2]

1. https://www.researchgate.net/publication/269949106_A_Study_on_the_Economic_Evaluation_for_a_Feasibility_Study_in_Molybdenum_Mineral_Processing
2. https://math24.net/probability-density-function.html

Cumulative distribution function (CDF)

- **Cumulative distribution function** (CDF) – $F_X(x)$ represent a cumulative representation of the PDF
- Describes the probability that the random variable is no larger than a given value
  - The area under the PDF from $-\infty$ to x
  - $F_X(x) = \int_{-\infty}^{x} f_X(u)\, du$

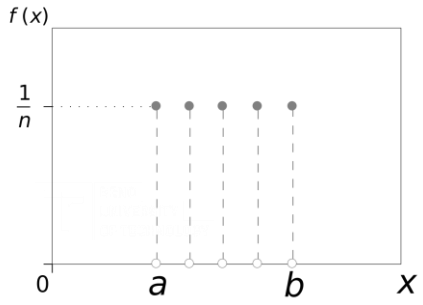


PDF and corresponding CDF [1]

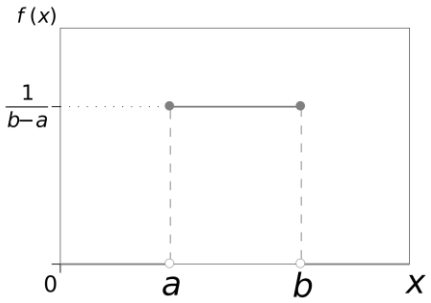
1. https://en.wikipedia.org/wiki/Probability_distribution

Uniform distribution

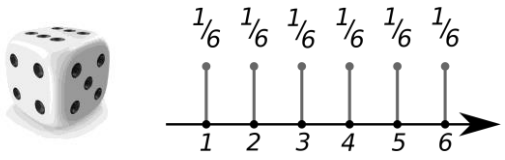
- **Uniform** distribution – all hypotheses will occur with **equal probability**
- **Discrete space** – a finite number of values has equal probability
  - $p(x) = \frac{1}{n}$ $\quad n = b - a + 1$ $\quad \mathrm{k} \in \{a, a+1, \cdots, b-1, b\}$ $\quad b \geq a$
- **Continuous space** – all intervals of the same length within the bounds are equally probable
  - $f(x) = \begin{cases} \frac{1}{b-a} \text{ for } x \epsilon [a, b] \\ 0 \text{ otherwise} \end{cases}$ $\quad -\infty < a < b < \infty$
- Examples
  - Rolling the dice
  - **Initial guess** in the **global localization** task



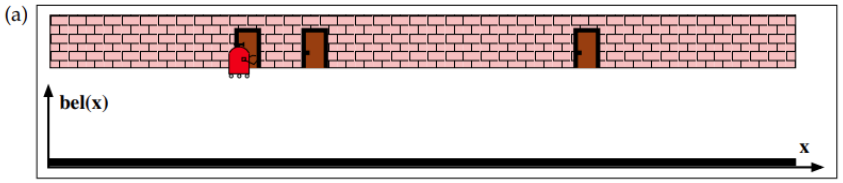
PMF of the discrete uniform distribution example [1]



PDF of the continuous uniform distribution example [2]



Dice rolling – typical discrete uniform distribution

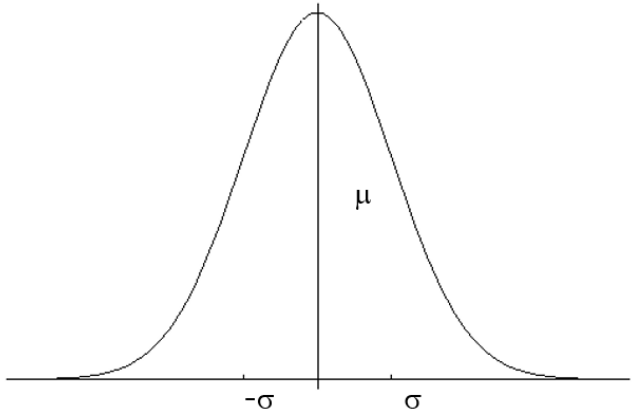


Initial guess with the uniform distribution [3]

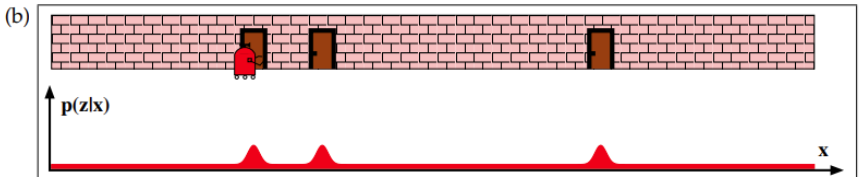
1. https://en.wikipedia.org/wiki/Discrete_uniform_distribution
2. https://en.wikipedia.org/wiki/Continuous_uniform_distribution
3. http://www.probabilistic-robotics.org/

Normal distribution

- **Normal** distribution = **Gaussian** distribution – type of **continuous** probability distribution
- Important in natural sciences – measurement errors caused by many independent processes posses nearly normal distribution
- Notation: $p(x) \sim \mathcal{N}(x; \mu, \sigma^2)$
- PDF: $\boxed{f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}}$
- Parametric distribution defined by:
  - **Mean** (expectation) μ
  - **Standard deviation** σ or **variance** σ^2
- Examples
  - Natural random processes
  - Approximate distribution of many **sensors in robotics**



PDF of normal distribution [1]

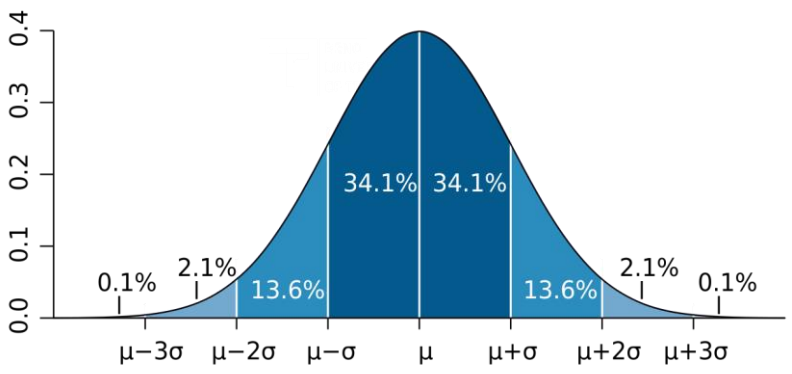


Sensor measurements are often near normal [1]

1. http://www.probabilistic-robotics.org/

Normal distribution – probabilities

- In the measurement field the uncertainty is often expresses by an interval of possible values occurring with certain probability defined via standard deviation σ
- variable $=$ value $\pm$ inverval typically meaning 1σ interval
- Well known probabilities for the normal distribution
  - $P(\mu - \sigma \leq X \leq \mu + \sigma) = 68{,}3\ \%$
  - $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) = 95{,}4\ \%$
  - $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) = 99{,}7\ \%$

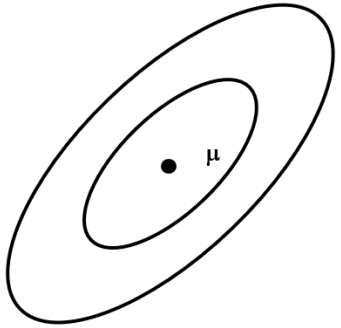


PDF of normal distribution with probabilities for various intervals defined with σ [1]

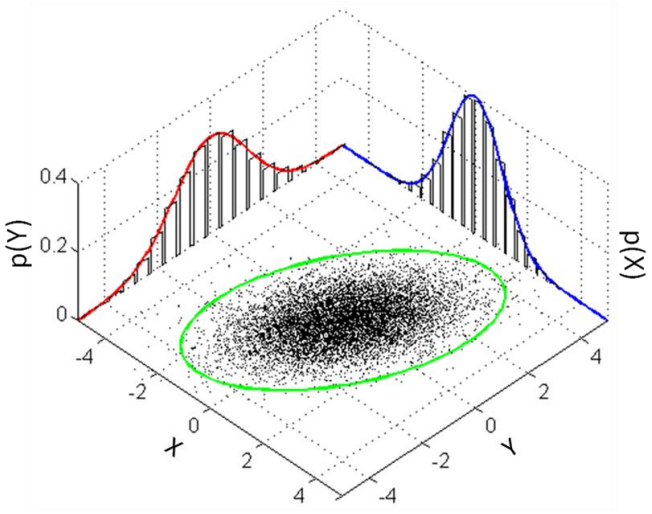
1. https://en.wikipedia.org/wiki/Normal_distribution

Multivariate normal distribution

- Normal distribution over vectors is called ***multivariate*** (x is a vector)
- Often used in localization algorithms when the space vector is multidimensional
- Notation: $p(x) \sim \mathcal{N}(x; \mu, \Sigma)$
- PDF: $p(x) = \det(2\pi\Sigma)^{-\frac{1}{2}} \exp\{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\}$
- Parameters:
  - **Mean vector** (expectation vector) μ
  - **Covariance matrix** Σ



[1]

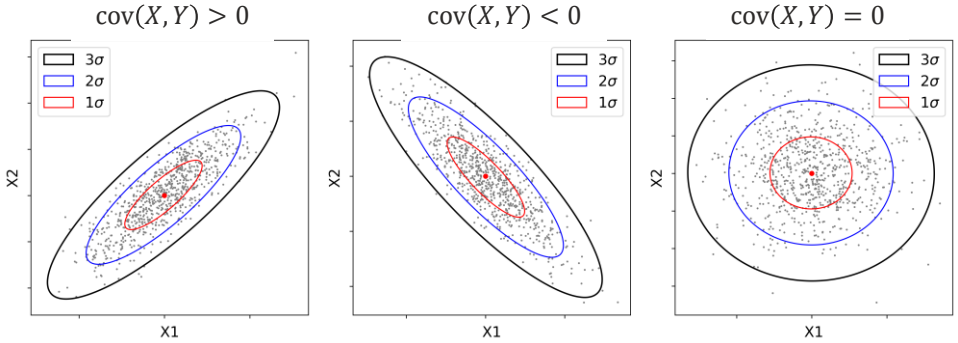


Normal distribution of two variables is often illustrated with ellipse in 2D space [2]

1. http://www.probabilistic-robotics.org/
2. https://en.wikipedia.org/wiki/Multivariate_normal_distribution

Covariance and covariance matrix

- ***Covariance*** is a measure of **the joint variability of two random variables**
  - The mean of the product of X and Y random variable deviations from their means
  - $\mathrm{cov}(X,Y) = E[(X - E[X])(Y - E[Y])]$ expectation $E[X] \approx \mathrm{mean}(X)$
  - Alternative notations: σ_{XY} or $\sigma(X,Y)$
  - **Covariance with itself** is **variance**: $\mathrm{cov}(X,X) = \mathrm{var}(X) \equiv \sigma^2(X)$
  - Zero covariance means **independent variables**
  - Normalized covariance → **correlation**
- ***Covariance matrix*** contains covariance between each pair of elements
  - Matrix dimension: dimensionality of the state x squared
  - Key matrix properties: square, symmetric, quadratic
  - $$\boxed{\Sigma = \begin{pmatrix} \sigma_1^2 & \cdots & \sigma_{1n} \\ \vdots & \ddots & \vdots \\ \sigma_{n1} & \cdots & \sigma_n^2 \end{pmatrix}}$$
  - **Main diagonal contains variances**



The covariance sign and the „shape“ of data/ellipses [1]

```
process_noise_covariance: [0.05, 0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,
                           0,    0.05, 0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,
                           0,    0,    0.06, 0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,
                           0,    0,    0,    0.03, 0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,
                           0,    0,    0,    0,    0.03, 0,    0,    0,    0,    0,    0,    0,    0,    0,    0,
                           0,    0,    0,    0,    0,    0.06, 0,    0,    0,    0,    0,    0,    0,    0,    0,
                           0,    0,    0,    0,    0,    0,    0.025, 0,    0,    0,    0,    0,    0,    0,    0,
                           0,    0,    0,    0,    0,    0,    0,    0.025, 0,    0,    0,    0,    0,    0,    0,
                           0,    0,    0,    0,    0,    0,    0,    0,    0.04, 0,    0,    0,    0,    0,    0,
                           0,    0,    0,    0,    0,    0,    0,    0,    0,    0.01, 0,    0,    0,    0,    0,
                           0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0.01, 0,    0,    0,    0,
                           0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0.02, 0,    0,    0,
                           0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0.01, 0,    0,
                           0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0.01, 0,
                           0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0,    0.015]
```

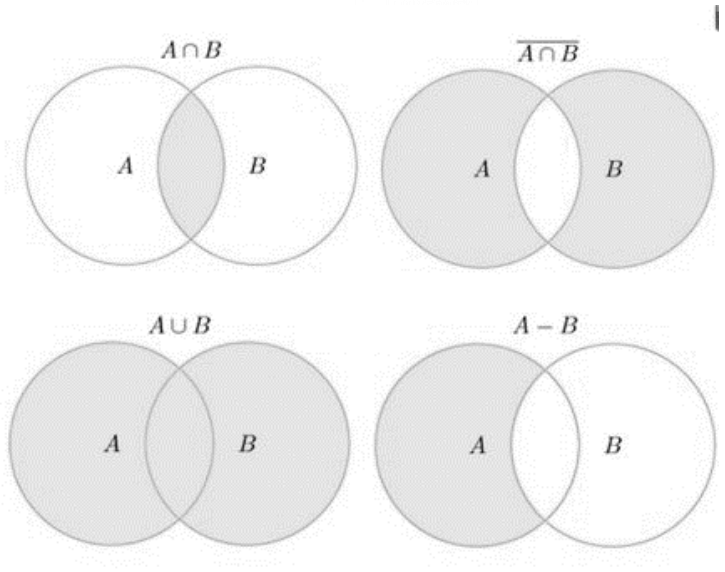
Covariance matrix describing noise characteristics of a process (robot motion) [2]

1. https://geostatisticslessons.com/lessons/errorellipses
2. https://answers.ros.org/question/353624/what-is-the-default-noise-parameters-in-sensor-inputs-in-robot_localization/

Combining probabilities

- The probabilities can be combined in different ways according to the phenomenon we are describing

Event	Probability
A	$P(A) \in [0,1]$
not A	$P(A^{\complement}) = 1 - P(A)$
A or B	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $P(A \cup B) = P(A) + P(B)$ if A and B are mutually exclusive
A and B	$P(A \cap B) = P(A\|B)P(B) = P(B\|A)P(A)$ $P(A \cap B) = P(A)P(B)$ if A and B are independent
A given B	$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B\|A)P(A)}{P(B)}$



- The most important event in the **probabilistic robotics** is **A given B**

Bayes' theorem

The theorem and its application in robotics

Bayes' theorem

- Statistical theory based on the Bayesian interpretation of probability
- **Probability** is based on **prior knowledge** of an event and related **evidence**
- The probability expresses a degree of **belief** in an event

$$\boxed{P(H|E) = \frac{P(E|H)P(H)}{P(E)}} = \frac{P(E|H)P(H)}{P(E|H)P(H) + P(E|\neg H)P(\neg H)} = \eta P(E|H)P(H)$$

- H **hypothesis** whose probability may be affected by evidence
- E **evidence** (observation) – new data that were not used in computing the prior probability
- $P(H)$ **prior** probability – the estimate of the probability of the hypothesis H before the data E
- $P(H|E)$ **posterior** probability – is the probability of H after E is observed
- $P(E|H)$ **likelihood** – probability of observing E given H
- $P(E)$ model evidence, probability of seeing evidence independently on H

Recommended resource: https://www.youtube.com/watch?v=HZGCoVF3YvM

Bayes' theorem

Likelihood

Posterior probability

Prior probability

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$

Model evidence

Bayes' theorem

Sensor model

New estimate

Old estimate

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$

Normalization

Bayes filter

- **Bayes filter** – standard approach for the **state estimation** in robotics
- General algorithm for calculating **belief** from **measurement** z and **control data** u
- **Recursive**, includes the previous **state** x_{t-1}
- Each period includes **prediction** and **correction** step
- Bayes filter forms the basis for widely used algorithms:
  - **Kalman filter** – Gaussian filter, belief is represented via normal distribution
  - **Particle filter** – Monte Carlo filter, belief is represented by a set of particles

Prediction (control update)

Algorithm Bayes_filter$(bel(x_{t-1}), u_t, z_t)$**:**

for all x_t do

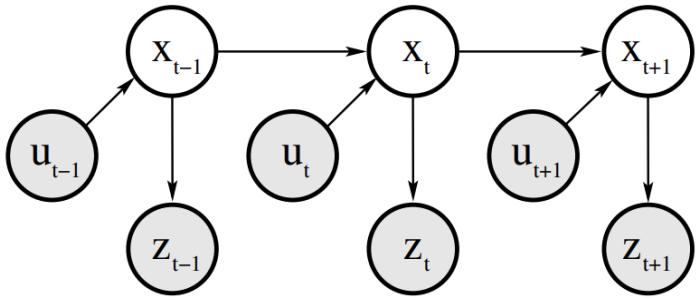
$\overline{bel}(x_t) = \int p(x_t \mid u_t, x_{t-1})\, bel(x_{t-1})\, dx$

$bel(x_t) = \eta\, p(z_t \mid x_t)\, \overline{bel}(x_t)$

endfor

return $bel(x_t)$

Correction (measurement update)



The evolution of controls, states and measurements [1]

1. http://www.probabilistic-robotics.org/

Bayes filter

Prior (initial) belief $\overline{bel}(x_0)$

Measurement $p(z_1|x_1)$

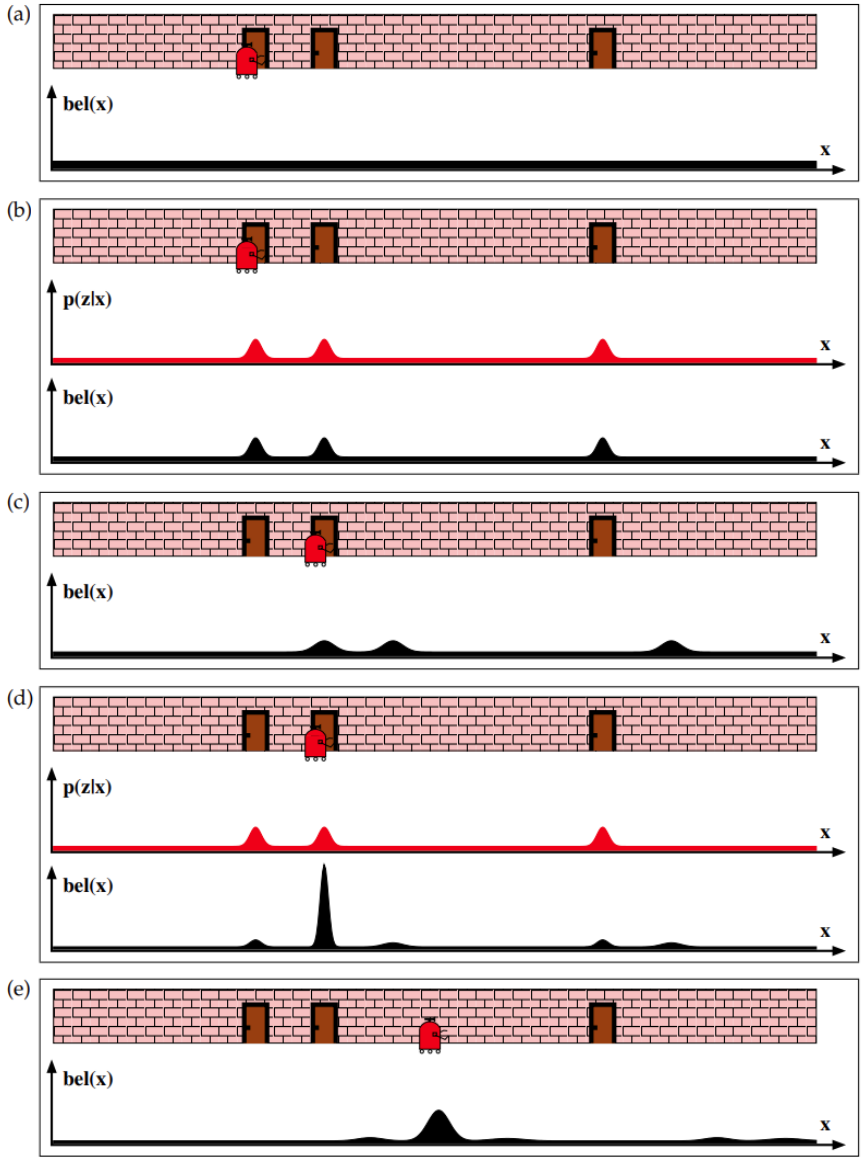
Correction, posterior belief $bel(x_1)$

Prediction, posterior belief $\overline{bel}(x_2)$

Measurement $p(z_2|x_2)$

Correction, posterior belief $bel(x_2)$

Prediction, posterior belief $\overline{bel}(x_3)$



[1]

Filter iteration 1

Filter iteration 2

Filter iteration 3

1. http://www.probabilistic-robotics.org/

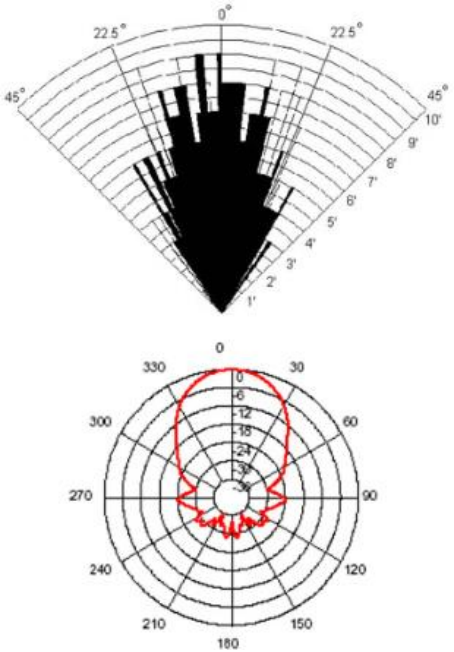
Sensors

Range and pose sensors

Range sensors

- Provide information about the **distance** and the (eventually) **azimuth** of the object with respect to robot's coordinate system
- The most common types:
  - **LiDAR** – narrow laser beam, time-of-flight
    - Mapping, localization, obstacle detection
    - 1D, 2D and 3D
  - **Ultrasonic** – wide „beam", time-of-flight
    - Obstacle detection
    - 1D only
  - **Radar** – wide „beam", time-of-flight, Doppler
    - Obstacle detection, velocity measurement
    - 1D (2D, 3D)

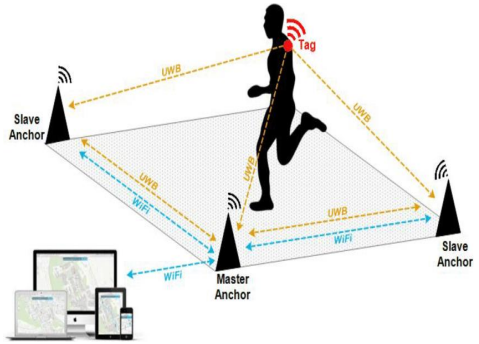
2D LiDAR scan



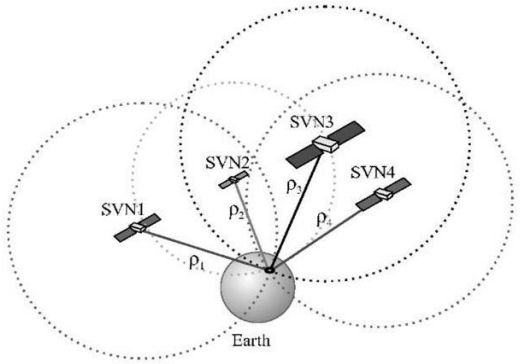
Ultrasonic sensor characteristics

External pose sensors

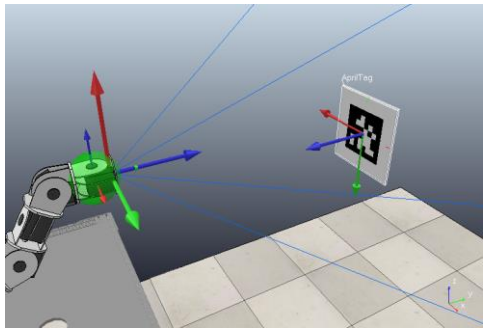
- Determines **position** or **orientation** or both position and orientation – **pose** within an external coordinate system
- The most common types:
  - **GNSS** – Global Navigation Satellite System
    - Global, world-wide localization
    - Position only, orientation – dual-antenna RTK systems
  - **Motion capture** and **beacon localization**
    - Local localization
    - Based on cameras and reflective markers or on radio beacons and receivers
  - **Fiducial markers**
    - Local localization
    - Relative pose of graphical marker and camera



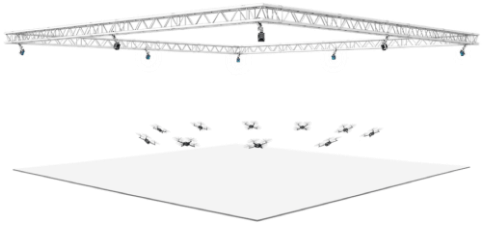
Beacon loclalization (UWB, BT, ...) [3]



GNSS [1]



Fiducial markers/tags [4]



Motion Capture [2]

1. http://what-when-how.com/
2. https://deva90sapmc8w.cloudfront.net/roboticsPrime17w_8-450h.png
3. https://www.researchgate.net/publication/336832935
4. https://haowensh.com/assets/img/2020/kdc/frame_confirmation.png

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